

Data Visualization Lab: Hands-On Practice

Welcome to the Data Visualization Practice Lab. For this assignment, you will be using a cleaned and preprocessed dataset titled **DV_Lab1_dataset.csv**. This synthetic dataset contains daily sales metrics for a fictional global tech retailer over the course of a year.

1. Dataset Overview (Data Dictionary)

Before you begin plotting, familiarize yourself with the columns available in the dataset:

Column Name	Data Type	Description
Date	Datetime	The date of the sales record (YYYY-MM-DD).
Region	Categorical	The store region (North, South, East, West).
Product_Category	Categorical	Category of items sold (Electronics, Software, Accessories).
Marketing_Spend	Numeric	Amount spent on marketing campaigns in USD.
Sales_Revenue	Numeric	Total revenue generated in USD.

Column Name	Data Type	Description
Customer_Age	Numeric	Average age of customers for that day.
Store_Traffic	Numeric	Total number of visitors to the store.

2. Plotting Assignments

Use Python libraries like matplotlib, seaborn, and pandas to complete the following tasks. Load your data using `pd.read_csv('GLA_Visualization_Lab_Data.csv')`.

Task 1: Bar Chart (Comparison)

- **Objective:** Compare the total *Sales_Revenue* across the four different *Regions*.
- **Instructions:** Group the data by Region, calculate the sum of *Sales_Revenue*, and plot a vertical bar chart. Ensure you add a title and label your axes.

Task 2: Line Chart (Trend over Time)

- **Objective:** Track the monthly trend of *Sales_Revenue*.
- **Instructions:** First, convert the 'Date' column to a datetime format and extract the month. Group the data by Month to get the total monthly sales, then plot a line chart with markers on each data point.

Task 3: Pie Chart (Composition)

- **Objective:** Show the market share of each *Product_Category* based on total revenue.
- **Instructions:** Group the data by *Product_Category* and sum the *Sales_Revenue*. Plot a pie chart showing the percentages (use `autopct`). Explode the slice with the highest revenue.

Task 4: Scatter Plot (Relationship)

- **Objective:** Investigate the correlation between *Marketing_Spend* (X-axis) and *Sales_Revenue* (Y-axis).

- **Instructions:** Create a scatter plot with these two continuous numerical variables. Add an alpha (transparency) parameter to easily visualize clustered points.

Task 5: Histogram (Distribution)

- **Objective:** Analyze the distribution of *Customer_Age* across the dataset.
- **Instructions:** Plot a histogram for the *Customer_Age* column. Experiment by setting bins=10 and then bins=25 to see how the distribution changes.

Task 6: Boxplot (Outliers & Spread)

- **Objective:** Compare the distribution and identify outliers of *Sales_Revenue* across different *Regions*.
- **Instructions:** Use Seaborn's `sns.boxplot()` with *Region* on the X-axis and *Sales_Revenue* on the Y-axis.

Task 7: Heatmap (Correlation)

- **Objective:** Find out which numerical features are most strongly correlated.
- **Instructions:** Select only the numeric columns (*Marketing_Spend*, *Sales_Revenue*, *Customer_Age*, *Store_Traffic*) and compute their correlation matrix using `df.corr()`. Plot this matrix using Seaborn's `sns.heatmap()` with annotations turned on.

Task 8: Bubble Chart (Multi-dimensional)

- **Objective:** Visualize the relationship between Marketing, Sales, and Traffic simultaneously.
- **Instructions:** Create a scatter plot where X is *Marketing_Spend*, Y is *Sales_Revenue*, and the size of the bubble (s parameter) is determined by *Store_Traffic*.

Task 9: Area Chart (Cumulative Trend)

- **Objective:** Show the cumulative *Sales_Revenue* over time stacked by *Product_Category*.
 - **Instructions:** Pivot your dataset so that dates are the index and product categories are the columns, filled with sales revenue. Use `plt.stackplot()` to visualize the volume over time.
-